

PRACTICAL 6 (A) - DNS AND HTTP

1. Devices Required

You need 6 devices:

Network Devices

1 Router — Router0

1 Switch — Switch0

End Devices

1 Server — Server0

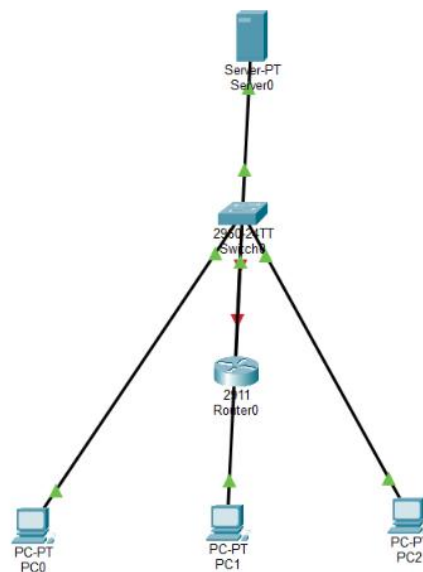
3 PCs — PC0, PC1, PC2

Connect the Copper Wires

Use Copper Straight-Through cable for all connections.

In Packet Tracer:

Connections ⚡ → **Copper Straight-Through**



From device	Port	To device	Port	Cable
Router0	GigabitEthernet 0/0	Switch0	GigabitEthernet 0/1	Copper Straight-Through
Server0	FastEthernet 0	Switch0	FastEthernet 0/1	Copper Straight-Through
PC0	FastEthernet 0	Switch0	FastEthernet 0/2	Copper Straight-Through
PC1	FastEthernet 0	Switch0	FastEthernet 0/3	Copper Straight-Through
PC2	FastEthernet 0	Switch0	FastEthernet 0/4	Copper Straight-Through

- **Configure Router0**

Click:

Router0 → Config → GigabitEthernet 0/0

Set:

IP Address: 10.0.0.1

Subnet Mask: 255.255.255.0

Make sure:

Port Status → On

So the router becomes:

Router0

IP: 10.0.0.1

- **Configure Server0**

Click:

Server0 → Desktop → IP Configuration

Enter:

IP Address: 10.0.0.2

Subnet Mask: 255.255.255.0

Default Gateway: 10.0.0.1

DNS Server: 10.0.0.2

So:

Server0

IP Address: 10.0.0.2

Gateway: 10.0.0.1

DNS: 10.0.0.2

- **Configure PC0**

Go to:

PC0 → Desktop → IP Configuration

Enter:

IP Address: 10.0.0.3

Subnet Mask: 255.255.255.0

Default Gateway: 10.0.0.1

DNS Server: 10.0.0.2

Configure PC1

Go to:

PC1 → Desktop → IP Configuration

Enter:

IP Address: 10.0.0.4

Subnet Mask: 255.255.255.0

Default Gateway: 10.0.0.1

DNS Server: 10.0.0.2

Configure PC2

Go to:

PC2 → Desktop → IP Configuration

Enter:

IP Address: 10.0.0.5

Subnet Mask: 255.255.255.0

Default Gateway: 10.0.0.1

DNS Server: 10.0.0.2

Configure HTTP on Server0

Go to:

Server0 → Services → HTTP

Make sure: HTTP: ON

Then click on (index.html) – edit

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Computer Networks Practical</title>
```

```
</head>
```

```
<body>
```

```
<h1>Welcome to My Website</h1>
```

```
<h2>Computer Networks Practical</h2>
```

```
<p>DNS and HTTP Server</p>
```

```
</body>
```

```
</html>
```

 (there might come a pop up or a warning like if you want to replace – click yes)

Configure DNS on Server0

Go to:

Server0 → Services → DNS

Turn:

DNS: ON

Add a DNS record:

Name	Address
google.com	10.0.0.2

So the DNS configuration is: google.com → 10.0.0.2

Click Add.

Check the Physical Connections

Before testing, all the links should eventually show green indicators.

Test HTTP Using the IP Address

On PC0:

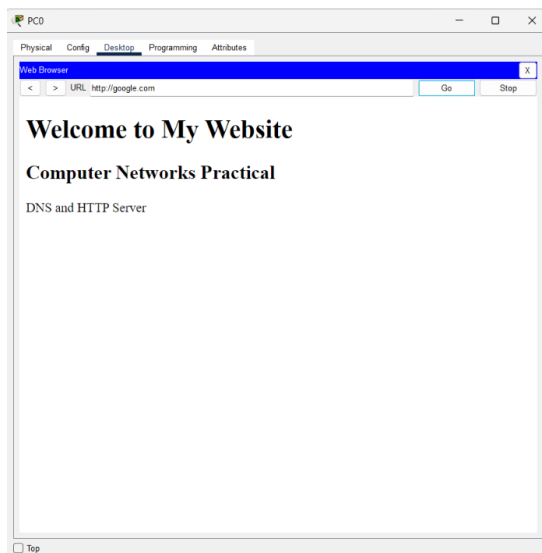
Desktop → Web Browser

Enter:

<http://google.com> or just google.com

Click Go.

Your webpage should appear:



This proves that the HTTP server is working.

(index.html content depend on what is given in the question)

PRACTICAL 6 (B) - DNS AND HTTP SERVER

Devices Required

You need:

1 Router — Router0

1 Switch — Switch0

1 Server — Server0

3 PCs — PC0, PC1, PC2

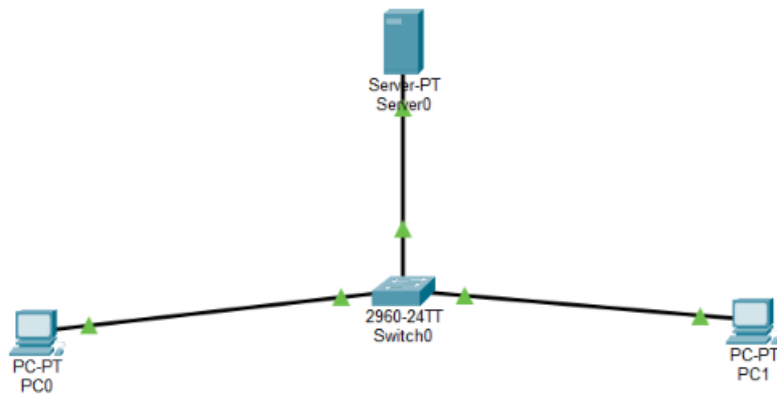
Connect the Copper Wires

Select:

Connections ⚡ → Copper Straight-Through

Make these connections:

From Device	Port	To Device	Port
Server0	FastEthernet 0	Switch0	FastEthernet 0/1
PC0	FastEthernet 0	Switch0	FastEthernet 0/2
PC1	FastEthernet 0	Switch0	FastEthernet 0/3



Wait until the connection indicators become **green**.

- **Configure Server0**

Go to:

Server0 → Desktop → IP Configuration

Enter:

IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: Leave blank

DNS Server: 192.168.1.2

Because there is **no router**, you do **not** need a default gateway.

- **Configure PC0**

Go to:

PC0 → Desktop → IP Configuration

Enter:

IP Address: 192.168.1.3

Subnet Mask: 255.255.255.0

Default Gateway: Leave blank

DNS Server: 192.168.1.2

- **Configure PC1**

Go to:

PC1 → Desktop → IP Configuration

Enter:

IP Address: 192.168.1.4

Subnet Mask: 255.255.255.0

Default Gateway: Leave blank

DNS Server: 192.168.1.2

- **Configure HTTP on Server0**

Go to:

Server0 → Services → HTTP

Make sure:

HTTP → ON

Open **index.html**.

Replace its contents with:

```
<!DOCTYPE html>

<html>

<head>

  <title>ABC.com</title>

</head>

<body>

  <h1>Welcome to ABC.com</h1>

  <h2>Computer Networks Practical</h2>

  <p>This website is hosted on Server0.</p>

</body>

</html>
```

- **Configure DNS on Server0**

Go to:

Server0 → Services → DNS

Turn:

DNS → ON

Add this record:

Name	Address
abc.com	192.168.1.2

Click **Add**.

Your DNS configuration is:

abc.com → 192.168.1.2

- **Test HTTP Using IP Address**

Go to:

PC0 → Desktop → Web Browser

Enter:

http://abc.com

Press **Enter**.

You should see:

Welcome to ABC.com

Computer Networks Practical

This website is hosted on Server0.